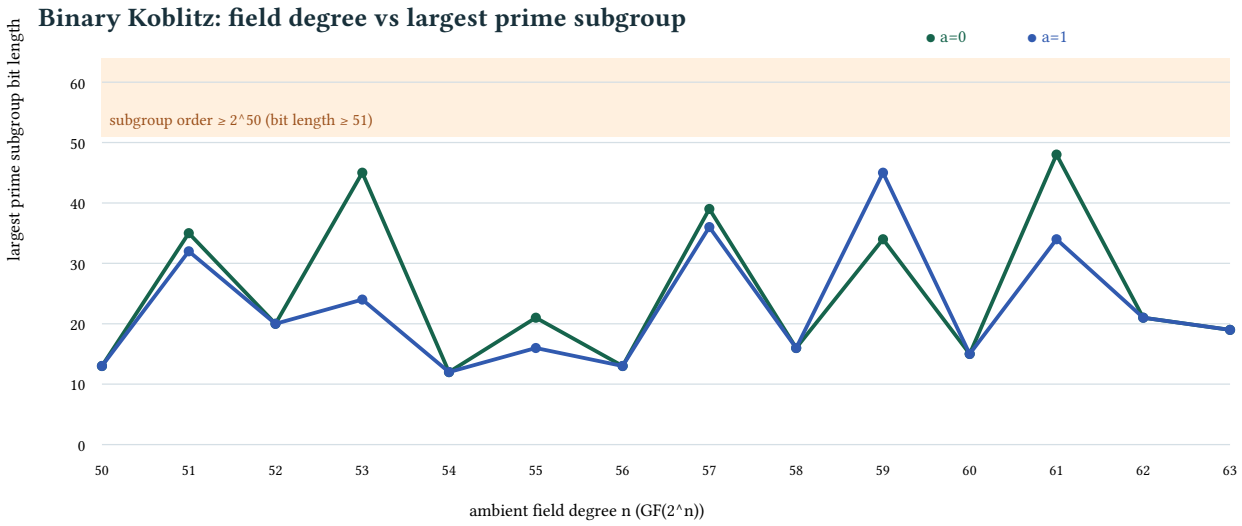
Koblitz field and subgroup sizes, n=50…63

Derived input-selection screen, 2026-10-17 UTC. The reviewed GF(2^53) Koblitz comparator has target prime r=21,044,858,204,113, about 2^44.26. Its field has 2^53 elements. These are different size coordinates.



For E_a: $y^2+xy=x^3+a\,x^2+1$ over F_2, direct point counts give trace −1 for a=0 and +1 for a=1. The exact recurrence $t_n = t_1\,t_{n-1} - 2\,t_{n-2}$, $t_0=2$, then gives group order over GF(2^n) as 2^n+1-t_n. Every order for n=50…63 was factored and reconstructed; every factor was checked prime.

Curve and field n	Largest r bits	Largest r
a=0, n=53	45	21,044,858,204,113
a=0, n=61	48	162,888,033,982,417
a=1, n=59	45	25,179,555,920,633

Neither standard binary Koblitz curve has a prime subgroup r≥2^50 at any field degree n=50…63. The reviewed n=53 result remains a complete 45-bit-subgroup comparator for its exact curve, not a 50-bit-subgroup row.

Higher field degrees with certified prime divisors

A bounded factor search at n=64…80 found three prime divisors in the requested 2^50…2^63 interval. Each is independently verified prime below 2^64 and divides the exact group order. Empty search rows are inconclusive.

Curve and field n	$\log_2$ r	Certified r
a=0, n=71	52.29	5,513,228,015,079,457
a=0, n=73	56.26	86,020,738,150,056,119
a=1, n=73	62.22	5,384,682,420,498,870,703

All three rows now have explicit prime-order generators from cofactor projection, checked in the source arithmetic and independently in Sage. Each Sage replay checks irreducibility, primality, point validity and rG=O. These are group fixtures; target panels, factor bases and a complete IC pipeline remain to be measured. No index-calculus timing or rho comparison was made here.

Data: research/icperf_koblitz_size_screen_20261017/orders.json and research/icperf_koblitz_subgroup_candidates_20261017/candidates.json. Fixture and Sage replay: research/icperf_koblitz_subgroup_fixtures_20261017/.